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Challenges

You can view the challenge statements and submit your abstract.

Note: There is no restriction on the number of abstracts a team may submit. Teams are permitted to submit multiple abstracts across different challenge statements.

Domain

S.No	Challenge title	Challenge statement	Domain	Action
1	Automation from Process Data sheet to Mechanical Data sheet	Create an AI-assisted proof of concept for one equipment category that converts a structured or semi-structured Process Data Sheet into a draft Mechanical Data Sheet using an approved template and engineering rules. The POC should extract key parameters, normalize units, identify missing or conflicting information, and link every populated field to its source. Users must review and correct extracted values before approving the MDS. The system should then compare sample vendor offers against the approved MDS, enquiry requirements, material specifications, and selected code rules. It should classify each requirement as compliant, deviating, unclear, or missing; explain the comparison; and generate clarification questions. Finally, it should prepare a draft deviation register and Technical Bid Evaluation with configurable scoring and an auditable	Manufacturing	Ask e Subn

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		synthetic, or mentor-approved documents.		
2	Patient Specific 3D Anatomical Reconstruction from DICOM Imaging Data	The challenge is to develop a reliable method for reconstructing a patient-specific 3D anatomical model from medical imaging data available in DICOM format. The solution should be capable of reading a sequence of DICOM image slices, interpreting the associated metadata, and maintaining the correct spatial relationship between individual slices during reconstruction. Variations in slice orientation, spacing, thickness, image resolution, patient movement and acquisition quality should be appropriately considered so that they do not significantly affect the final anatomical representation. The reconstructed 3D volume should support identification and separation of relevant anatomical structures present within the selected region. The solution should also be capable of refining the reconstructed structures and generating an anatomically meaningful 3D mesh that closely represents the original patient anatomy. Particular importance should be given to alignment accuracy, structural continuity, anatomical consistency and preservation of clinically relevant features. The developed approach should further provide an interactive visualization of the reconstructed anatomical	MedTech	

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		of important structural or tissue-related characteristics wherever applicable. The overall objective is to achieve a consistent, accurate and usable 3D anatomical representation that can support medical visualization, research analysis, diagnostic assessment, treatment planning and surgical planning.		
3	Automated Asset Recognition and Intelligent 3D Model Generation from Laser Scan Data	Develop and demonstrate a solution that turns laser-scan point-cloud data into engineering information with less manual effort. Participants may choose a plant area, asset types, data sources, tools, AI methods, engineering libraries, and form of POC. The solution should identify selected objects, describe what they are, show where they are located, and create or support an engineering-ready three-dimensional model. Teams may also use drawings, equipment lists, photographs, or information to improve the result. The system should show confidence, highlight uncertain or missing areas, and allow an engineer to review and correct its output. Participants should test the solution using representative scan conditions, including clutter, noise, partial visibility, or incomplete data where appropriate. Results should be compared with a trusted reference using measures such as recognition accuracy, location and size	Manufacturing	

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		processing time, and manual effort saved.		
4	Object-Detection Model for Battery Assembly & Maintenance	Train an object-detection/vision model that verifies battery component orientation and assembly correctness from images/video, flags misorientation or missing steps in real time, and guides technicians through the correct sequence. Deploy in a US-based/ITAR-compliant manner with on-prem inference, and log each verified step for traceability.	Manufacturing	Ask e Subn
5	Personalized Streaming Without the Bandwidth Cost	Design a solution that delivers a personalised media experience (e.g., adaptive video quality, personalised content assembly, or targeted ad insertion) while minimising bandwidth consumption under constrained or variable network conditions. Approaches may include perceptual or content-aware compression, edge caching, predictive pre-fetching, or AI-driven adaptive encoding that allocates bandwidth to what matters most to the viewer. The primary metric is objective perceptual quality per megabyte delivered, measured against standard adaptive bitrate streaming (HLS or DASH) as the baseline on a common test clip set and network degradation profile, with rebuffering ratio and startup latency as secondary metrics.	Digital Platforms	Ask e Subn

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		Develop an AI-driven closed-loop tuning POC for one simulated plant: a bidirectional EV fast-charger DC-link voltage controller under step-load and grid-voltage disturbances. Use a provided or reproducible MATLAB/Simulink or Python plant model with fixed voltage, current, stability, and switching constraints. The learning or optimization method shall tune PI gains online within certified bounds, fall back to a safe baseline controller when confidence or constraints fail, and log every gain change. Compare against a manually tuned PI controller over identical operating points, parameter variations, sensor noise, and disturbances. No connection to live high-voltage hardware is required.		
6	AI-Driven Closed-Loop Control Optimization		Energy	Ask e Subn
7	AI enabled Conceptual Design & Costing for Pharmaceutical Clients	Create an AI-assisted proof of concept for one pharmaceutical facility area, such as a production block or utility zone. The POC should accept a structured brief containing room requirements, major equipment, capacities, process relationships, material and personnel flows, selected design rules, and cost parameters. It should generate two or three conceptual layout alternatives, rank them using transparent criteria, and explain key trade-offs. The selected alternative must be represented in a lightweight 3D Building Information Modeling (BIM) model showing rooms, major	Manufacturing	Ask e Subn

relationships. The system should perform basic rule checks for zoning, adjacency, flow separation, safety, maintainability, and space allocation, then produce an indicative Total Investment Cost range with traceable assumptions. Use synthetic, sanitized, or mentor-approved reference data.

8	AI-Driven Engineering Change Management for Resilient Material and Sourcing Decisions	Develop an AI-assisted proof of concept for material-impact assessment and phase-in/phase-out planning for one defined engineering-change scenario. Using synthetic, sanitized, data, the POC should ingest a change record, affected bill-of-material items, effective date, inventory balances, open purchase orders, demand, lead times, supplier information, and restricted-purchasing indicators from simulated ECM and SAP-like sources. It should identify impacted old and new materials, calculate projected inventory exposure, detect likely shortages or excess stock, and highlight sourcing constraints. A rules-based workflow should recommend actions such as consume existing stock, revise an effective date, expedite or defer an order, seek an alternate source, or request planner review. AI may summarize risks, prioritize exceptions, and explain recommendations. The dashboard should show change	Manufacturing	

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		decisions, and an auditable action log.		
9	Enterprise Digital Twin of Factory	Design and demonstrate an innovative proof of concept for a factory digital twin that addresses a defined manufacturing need. Teams may select an appropriate process, production area, asset group, material flow, utility system, or operational decision as the focus. The solution should establish a meaningful relationship between the factory context and its virtual representation, keep the twin current and clearly demonstrate how users gain insight or evaluate alternatives. Teams may choose the modeling approach, data sources, sensing methods, hardware, software, analytics, visualization, simulation, artificial intelligence, and interaction model. The POC should explain its assumptions, boundaries, data requirements, intended users, and measurable success criteria. It should show how the concept could evolve beyond the demonstration without claiming plant-wide completeness. The emphasis is on originality, usefulness, reasoning, and evidence that the proposed twin solves the selected problem.	Manufacturing	Ask e Subn
10	Leveraging AI in CAD cleansing activities	Demonstrate a proof of concept that improves a defined CAD cleansing activity or workflow. Teams may choose the CAD	Manufacturing	Ask e

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		quality problems, customer rules, lifecycle stage, and downstream use. The solution should analyze sample models or extracted information, identify or prioritize issues, and support correction, guidance, or review. Participants may combine configurable checks, geometric analysis, metadata reasoning, artificial intelligence, automation, human-in-the-loop interaction, or other justified approaches. The POC should show how rules or knowledge can be adapted for more than one scenario without claiming universal support for all CAD systems. It must distinguish safe automatic actions from changes requiring engineer approval, retain traceability to the original model, and provide evidence that design intent is not altered. Teams should define success measures such as detection quality, correction accuracy, effort reduction, consistency, or downstream readiness.		
11	Non-Contact Crack Detection and Digital Twin Generation for Dies and Moulds	A XY positioning system tracks each crack location while the next crack location is being captured. The solution should identify defect geometry, generate an accurate spatial profile, and create a digital representation that can be used for inspection, analytics, maintenance planning, or automated repair applications. The proposed system may leverage computer vision,	Manufacturing	

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		such as laser, artificial intelligence, image processing, robotics, motion control, or a combination of these approaches. The solution should demonstrate the ability to accurately capture defect characteristics and support traceability through digital coordinate extraction or digital twin generation.		
12	Transforming Legacy Factories into Intelligent Manufacturing Facilities	Develop a smart-manufacturing modernization proof of concept for one defined production line, cell, or utility area using synthetic, historical, or mentor-approved plant data. The POC should document the current process, equipment, data sources, material flow, major losses, and operational constraints. It should define baseline measures for selected outcomes such as availability, cycle time, throughput, quality, energy, or safety. Teams should choose one or two priority use cases, for example condition monitoring, downtime classification, energy monitoring, production visibility, or bottleneck detection, and demonstrate them using simulated or low-cost sensor data. A dashboard should present current status, trends, exceptions, and the calculated business effect of identified losses. The solution should produce a future-state architecture and prioritized roadmap showing required sensors, connectivity, data flow,	Manufacturing	

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		integration points, estimated effort, benefits, dependencies, and risks.		
13	Non-Invasive Knee Movement Tracking for Gait Assessment	Design and validate a non-invasive system that measures six-degree-of-freedom knee kinematics during functional movement, and present the output in a form a clinician can interpret. Teams may combine wearable inertial measurement units, depth or multi-camera vision, flexible strain sensing, or other non-invasive modalities, together with a sensor-fusion algorithm that resolves rotation and translation simultaneously. The system must operate outside a dedicated gait laboratory, tolerate soft-tissue movement, and remain usable by a single operator. Teams are expected to define a movement protocol (for example gait, sit-to-stand, stair descent and a controlled pivot), acquire data from healthy volunteers, and validate their output against an accepted reference such as an optical motion-capture system or an instrumented rig. Accuracy targets: rotational error within 5 degrees and translational error within 5 mm root-mean-square against the reference, sustained across the full movement cycle. Implantable or in-vivo sensing is explicitly out of scope; ligament-level strain measurement may be discussed only as future work.	MedTech	

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		Develop and demonstrate a working physical POC for a ceiling-mounted air-moving system without exposed rotating blades. Participants may choose the shape, airflow principle, internal mechanism, materials, controls, manufacturing method, and appearance. The prototype should create useful airflow in a chosen room or test area and show how the design improves safety or another important user need. Teams should compare it with a suitable conventional fan or reference using measures such as air movement, comfort coverage, noise, power use, size, weight, cost, vibration, cleaning, maintenance, and installation. Calculations, airflow simulation, or digital models may support the design but cannot replace the physical prototype. The solution should include protection against contact with moving parts, overheating, electrical faults, loose mounting, and other relevant risks.		
14	Reimagining Ceiling Airflow Designing a Safe Bladeless Ceiling Fan		Energy	Ask e Subn
15	AI Tool for Serviceability Evaluation- Via Design Data	Develop an AI-assisted proof of concept that imports a selected CAD or 3D product assembly, identifies serviceable components and their relationships, and generates an explainable serviceability score using configurable engineering criteria. The POC should evaluate access paths, disassembly steps, tool and hand clearance, fastener	Manufacturing	Ask e Subn

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		replaceability, estimated repair time, ergonomics, modularity, and safety constraints. It should highlight obstructing parts on the 3D model, show how each criterion contributes to the overall and subsystem scores, compare design revisions, and recommend practical improvements. The system should preserve traceability to the source geometry, bill of materials, rules, and any historical service data used. Validation should use approved designs or controlled test assemblies and compare the generated assessment with expert review. A hardware-assisted validation module may use a depth camera, AR headset, or instrumented physical mock-up to capture real access, reach, visibility, and tool-clearance evidence and compare it with the CAD-based prediction.		
16	Digital Path Teaching and Localization System for Variable Manufacturing Environments	Develop a wireless digital pen capable of recording the spatial coordinates of user-selected points on a planar work surface through a button-triggered input. The recorded coordinates shall be communicated to an XY positioning stage, which will automatically move to each stored location for point tracing or verification.	Manufacturing	Ask e Subn
17	Spatial-Analysis-to-BOM Generator for LightingDesign	Build a prototype that analyzes a room's spatial layout (image/scan/floorplan), detects dimensions, fixture positions,	Manufacturing	Ask e

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		automatically generates a validated lighting/controls BOM from the Nexus catalog. Computer vision extracts spatial features; a rules-plus-LLM layer selects compatible products, checks coverage, and explains each selection.		
18	AI-Powered Multimodal Surgical Assistant for Real-Time Operating Room Guidance	Develop a nonclinical conversational assistant for one simulated operating-theatre workflow: the pre-incision safety checklist. Using scripted scenarios, synthetic patient data, and one modality—full-duplex speech—the assistant shall recognize checklist items, retrieve approved protocol content, track completion, identify omissions or contradictory responses, and provide concise prompts. It must handle interruptions, overlapping speech, noise, and uncertain recognition; never make diagnosis or autonomous treatment decisions; and require the designated clinician to confirm every checklist completion. Compare with a static digital checklist in controlled simulations.	MedTech	Ask e Subn
19	Aqua Birth - Smart Water Birth System with Underwater Wireless Communication for Maternal & Fetal Monitoring	Design and build a working demonstration of a waterproof, cable-free monitoring system for water birth. Small wearable or floating sensors on the mother should keep measuring her heart rate, blood pressure, body temperature, blood oxygen level, and the timing and strength of her	MedTech	Ask e Subn

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should follow the baby's heart rate, movement and any early sign of distress. All this data must travel through the water to a bedside screen without wires, and appear almost immediately so that staff can act in time. Teams will have to find a way of sending signals through water. The system must be fully sealed, run for the length of a long labour on a small battery, stay comfortable while the mother moves, and raise a clear alarm whenever a reading becomes unsafe.

20	Validating Trust in the Software-Defined Vehicle	Propose and demonstrate a testing or validation approach — using simulation, virtual hardware-in-the-loop, or AI-driven test generation — that detects a software regression or safety-critical fault in a vehicle feature (e.g., ADAS, battery management, or an infotainment-to-safety interaction) before deployment. The approach must be usable within a compressed OTA release cycle measured in days, and teams must clearly articulate how test cases are generated, prioritised, and assessed for coverage. Teams may choose to work with vehicle simulators (e.g., CARLA), synthetic sensor data, or mocked ECU environments.	Mobility	Ask e Subn
21	AI Reasoning model for an efficient commercial	Develop and demonstrate a system that improves the efficiency and reliability of a commercial refrigerator, freezer,	Energy	Ask e

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	system	choose the equipment, users, operating conditions, data sources, hardware, software, and technical approach. The solution should understand changing cooling needs and help decide when actions such as cooling or defrosting may be required. It may also identify unusual behaviour linked to the compressor, power supply, airflow, vibration, doors, temperature, or other conditions. Teams should explain how the system handles missing or incorrect data, changing loads, sensor problems, and situations where it is uncertain. The POC should compare its performance with the normal operating method using measures such as energy use, temperature stability, defrost effectiveness, warning time, false alarms, and equipment health.		
22	AI Agents for Resilient Supply Chain	Develop a multi-agent supply-chain POC for a defined three-echelon network facing one disruption: a seven-day shutdown of a critical supplier. Use a supplied or reproducible dataset containing demand, inventory, capacity, lead time, transport cost, and service constraints. Demand, inventory, supplier-risk, and logistics agents may propose actions, but a coordinator must reconcile them using explicit business constraints and a mathematical optimization baseline. The system shall	Manufacturing	

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		alternate sourcing, allocation, or rescheduling; explain trade-offs; and require planner approval. Compare the agent solution with reorder rules and optimization-only planning over identical scenarios.		
23	Smart Visual Planning Tool for Lift Modernization	Design and demonstrate a proof of concept that helps users plan and visualize the modernization of an existing lift. The system should accept photographs or short videos of a lift cabin, entrance, or lobby together with basic site measurements and user preferences. It should identify the main visible areas, such as doors, panels, lighting, controls, walls, flooring, and display units, and suggest suitable upgrade options from a provided product list. The tool should create at least two realistic before-and-after visualizations while keeping the size, viewpoint, surrounding structure, and unchanged parts consistent with the original scene. It should clearly show which items are proposed, why they were selected, and whether any important information is missing. The system should also prepare a simple draft proposal containing the selected options, assumptions, reference images, and approximate work scope. Users must be able to edit or reject suggestions, and a qualified person must review all	Manufacturing	

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with a customer.

24	Novel pinless/markerless registration methods for navigation and surgical robotics	Develop a markerless registration proof of concept for one orthopedic anatomy and workflow, using a synthetic bone model, physical phantom, or approved dataset. The POC should create a three-dimensional reference model from CT or equivalent imaging and acquire intraoperative-like surface or depth observations using stereo vision, structured light, depth sensing, or another justified modality. It should automatically identify usable geometric features, estimate the rigid transformation between observed anatomy and the reference model, and display the aligned result. The system must calculate registration error and confidence, detect insufficient coverage or loss of alignment, and allow re-registration when required. Evaluation should include different viewpoints, partial occlusion, sensor noise, and controlled phantom movement. Teams should compare the method with a point-based or surface-based baseline. Instrument tracking can be demonstrated as an extension.	MedTech	

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25	AI-Enabled Suspension Digital Twin with Sensor Fusion for Ride Comfort Prediction	Develop an AI-assisted suspension digital-twin proof of concept that combines sensor fusion, road-state estimation, vehicle-response simulation, and ride-comfort prediction for a quarter-car or half-car model. Using simulated, public data, the POC should ingest body and wheel acceleration, suspension travel, wheel speed, vehicle speed, and inertial measurements. It should synchronize signals, filter noise, detect dropouts, and estimate selected road and suspension states using a justified fusion method. These estimates should drive a physics-based model and an AI/ML model that predicts an objective comfort score using an agreed vibration method. The dashboard should compare raw signals, fused states, simulated response, predicted comfort, confidence, and residual error across defined road, speed, load, and damping scenarios. For a semi-active case, the system may compare predefined damping settings and recommend the best evaluated option.	Mobility	Ask e Subn
26	AI-for-Design Generative Engineering Assistant	Develop a generative engineering assistant for one bounded product family: low-voltage motor-starter panels using a supplied component catalog and a versioned subset of IEC/UL design rules. The assistant shall convert a structured natural-language design brief into candidate component selections, sizing	Manufacturing	Ask e Subn

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layout, and an explained compliance report. All numeric sizing and rule checks must be executed by deterministic calculators or a rules engine; the language model may interpret intent and explain results but may not invent components, ratings, or standards. Ambiguous or missing requirements must trigger questions, and an engineer must approve the final draft.

27	Smart Digital Setup Assistance for Rapid and Error-Free Assembly Line Changeovers	Design and demonstrate an innovative proof of concept that makes manual assembly-line changeovers or equipment setup faster, more consistent, easier to learn, and less dependent on individual experience. Participants may select a representative process and define the users, operating context, form of guidance, information sources, interaction method, level of automation, and implementation platform. The solution should help operators access the right knowledge at the right time, recognize critical or missed activities where appropriate, and capture useful evidence about progress, delays, exceptions, quality, or learning needs. Teams are encouraged to question the current process rather than merely reproduce paper instructions on a screen. Approaches may use software, connected tools, physical interfaces, sensing,	Manufacturing	

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		intelligence, workflow redesign, or other justified combinations. The POC should accommodate changing products and procedures, intermittent connectivity, legitimate deviations, and different operator needs while maintaining safety, security, revision control, and respectful use of workforce data. Participants should define meaningful baseline and outcome measures and demonstrate how the concept could exchange information with relevant shop-floor systems. The goal is a practical and scalable improvement concept, not a prescribed device, workflow, or technology stack.		
28	Smart Precision Motion Platform for Ophthalmic Imaging	Develop and demonstrate a precision motion platform that positions an eye-imaging head around a seated patient. The platform should move along and around several axes, travel smoothly and repeatably, and settle with very little vibration. It must move between the left and the right eye on its own, along a planned path that always stays outside a defined no-go zone around the patient's face, and it should never require the patient to be repositioned. Position sensors should confirm the actual position at all times, detect slip or lost motion, and trigger a safe stop and recovery if anything goes wrong. Operators should also be able	MedTech	

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		alignment by hand through an intuitive control, with the speed reducing as the head comes closer to the eye.		
29	AI-Powered Leadless Cardiac Monitoring for Early Arrhythmia Detection	Design and demonstrate a proof of concept for continuous or repeated cardiac-rhythm monitoring outside a conventional multi-lead clinical setup. Teams may define the target users, monitoring context, rhythm events, sensing approach, device form, data sources, and interaction model. The solution may use electrical, optical, mechanical, acoustic, motion, or contextual signals, together with signal processing, artificial intelligence, or other justified methods. It should address signal quality, artifacts, uncertainty, personalization, and the difference between screening information and clinical diagnosis. Participants should show how potential abnormalities are detected or prioritized, how evidence is presented for review, and how the user is informed when data quality is insufficient.	MedTech	Ask e Subn
30	Multipurpose AI-Based Pedestrian Safety System Using RGB and Thermal Sensor Fusion	Develop and demonstrate a system that helps prevent collisions between pedestrians and moving equipment in a workplace. Participants may select the setting, users, hazards, information sources, hardware, software, and technical approach. The solution should detect or understand when a person may	Manufacturing	Ask e Subn

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		movement of the person and equipment, and provide a timely, clear warning or practical response. It should work under conditions that may include darkness, glare, dust, smoke, rain, blind spots, noise, or partial blockage. Teams should explain how the system handles missed information, false alarms, equipment failure, and situations where it is not confident. The warning should be easy to understand and should not distract or overwhelm the operator. The POC should be tested using safe and realistic scenarios with clear measures such as detection rate, missed cases, false alarms, warning time, range, and response speed. Any automatic control action must be demonstrated safely.		
31	Self Calibrating and Healing Medical Devices Using AI and Digital Twins	Build a working demonstration of a medical device that can look after its own calibration. Pick one or two device types and their sensors, for example pressure, flow, temperature, optical or ECG, and create a system that continuously checks whether the readings can still be trusted. The solution should spot slow drift early, tell real drift apart from ordinary noise and from changes in room temperature or humidity, and estimate how long a sensor has left before it must be corrected or replaced. Where the design allows, the device should correct itself automatically;	MedTech	

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		clearly assit on what to do and when. A digital twin, that is a software copy of how the device is expected to behave, can be used to compare expected against actual behaviour.		
32	AI-Vision Powered Smart Inspection System for High-Speed Consumer Electronics PCB Assembly	Develop an AI-Vision Smart Inspection System for consumer electronics PCB assembly that: (a) trains deep learning models (object detection + classification) on PCB defect image datasets to detect and classify common SMT defects (solder bridges, tombstoning, missing components, misalignment, insufficient solder), (b) demonstrates significant reduction in false-call rate compared to rule-based AOI baseline, (c) correlates defect data with upstream process parameters (solder paste volume from SPI, placement accuracy) to identify root causes, (d) provides a real-time defect analytics dashboard with Pareto analysis, trend monitoring, and SPC charts, and (e) proposes a closed-loop feedback mechanism to upstream equipment for process auto-correction.	Manufacturing	Ask e Subn
33	Real Time ADAS To Detect Parked Vehicle to overtake another vehicle.	Design and demonstrate an innovative proof of concept that improves awareness of hazards and uncertainty during highway overtaking, particularly when the driver or vehicle has a restricted line of sight. Participants may define the	Mobility	Ask e Subn

operating conditions, information sources, technical approach, system boundaries, and form of demonstration. Solutions may use onboard, cooperative, connected, infrastructure-based, simulated, or other justified methods, individually or in combination. The POC should interpret relevant traffic conditions, distinguish meaningful hazards from benign objects, consider vehicle motion and available reaction time, and communicate information in a way that supports rather than replaces responsible driver judgement. Teams should determine how their concept responds to occlusion, adverse weather, changing illumination, incomplete data, conflicting observations, and low-confidence situations. Testing should include representative normal, difficult, and failure scenarios, with clear evidence of what the solution can and cannot reliably address. Any physical actuation must be demonstrated only in a safe and controlled environment. The intended outcome is an explainable, testable concept that advances overtaking safety.

34	Sustainable Cooling Solutions for Next-Generation AI Data Centers	Develop and demonstrate a working physical POC that explores a sustainable way to cool high-density AI computing equipment. Participants may choose the use case, scale,	Energy

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materials, hardware, controls, monitoring approach, and testing. The prototype should create or use a heat load and show that the idea can remove or manage heat under representative conditions. Teams should decide which performance measures are most relevant, such as temperature control, energy use, water or fluid use, response time, reliability, cost, size, ease of maintenance, or scalability.

35	Current-State Packaging Data and Digital Readiness Assessment for PPWR Compliance Reporting	Develop and demonstrate a POC that helps an organization understand its packaging data and readiness for compliance and sustainability reporting. Participants may choose a product group, packaging types, data sources, assessment method, technology, and user experience. The solution should import sample packaging information, compare it with a defined set of requirements, identify missing or conflicting data, and show where supporting evidence is available. It should help users understand packaging material use, recyclability, recycled content, reduction opportunities, supplier information, or other selected factors. Teams should provide clear explanations rather than a simple pass or fail result and allow users to review or correct findings. The POC may use	Sustainability	

data and should not claim formal regulatory approval. Results should be measured through data completeness, accuracy, traceability, time saved, and usefulness of recommendations.

36	OTT streaming quality of service	Create a non-intrusive hardware-and-software proof of concept that monitors selected live or recorded OTT streams from the viewer perspective without modifying the source stream or player. The POC should capture screen output and audio, collect available network or playback telemetry, and identify black frames, frozen frames, startup delay, buffering events, playback interruptions, audio-video synchronization errors, and bitrate or resolution changes. It should correlate detected events with bandwidth, latency, packet loss, and device resource information when available. The system must calculate an explainable Quality of Experience score using configurable thresholds and show how each detected issue affects the score. Results should include timestamps, event duration, confidence, screenshots or clips, and possible contributing factors. Testing should cover a defined set of devices, resolutions, protocols, content samples, and controlled network conditions. The POC is for monitoring and	Digital Platforms	

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		teams; production-scale deployment, exhaustive protocol coverage, and automated service remediation are outside scope.		
37	Edge-AI Wearable for Continuous Remote Vital-Sign Monitoring	Design and build a low-cost, low-power wearable prototype that continuously captures at least one physiological signal (e.g., SpO2, ECG, heart rate, or motion) and performs on-device anomaly detection using an embedded AI/ML model. A functional hardware prototype is mandatory. The solution must operate correctly with intermittent or no network connectivity, buffering data locally and syncing opportunistically. The primary metric is detection latency, measured from the onset of the anomaly in the signal to the device raising it, with the false-positive rate over a continuous multi-hour run as the counterweight. Teams may use any combination of low-power microcontrollers, edge-AI accelerators, or TinyML frameworks, and must consider their power budget, sensor accuracy trade-offs, and a realistic caregiver alerting mechanism. Consideration should also be given to data privacy handling.	MedTech	Ask e Subn
38	Adaptive AI driven Medical Scan Analysis	Develop and demonstrate an AI-based medical scan support system that can adjust how it helps different doctors and clinical situations. Participants	MedTech	Ask e

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condition, user group, data source, and approach. The system could change the amount of detail, explanation, evidence, or guidance it provides based on the user's needs, without making assumptions about a doctor's ability. It should show findings, explain why they may matter, and state its confidence and limits. It should also recognize poor-quality or incomplete scans and ask for review rather than giving a misleading answer. Doctors must be able to question, correct, or reject the system's suggestions. Teams should test whether the doctor and AI working together perform better than either alone, using measures and representative users or simulations. The solution should protect patient privacy, reduce bias, keep a record of decisions, and remain a support tool rather than replace medical judgement.

39	3D mapping and segmentation using any autonomous mobile robot	The core challenge requires processing multi-sensor payload feeds (LiDAR, RGB cameras) to perform 3D mapping of the area in a GPS-denied environment . The base system must accurately reconstruct dense 3D environments and classify geometric features and physical assets (e.g., structural pillars, equipment, pathways, fences, humans etc.) without manual labeling. It should work for any	Robotics & Autonomous Systems	

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system. Bonus Capabilities: 1. Integrating full platform autonomy with a. active exploration (more points for exploratory than following a path) b. real-time obstacle avoidance 2. Building a dynamic, cloud-synchronized Digital Twin dashboard featuring a. asset identification b. automated structural change detection c. asset condition monitoring d. spatial querying

40	Predictive Safety on the Factory Floor	Design and build a low-cost, retrofit-friendly monitoring solution that uses sensor fusion (e.g., vibration, temperature, acoustic, or current draw) combined with an edge-AI or rule-based model to predict impending equipment failure or unsafe operating conditions on industrial machinery. A functional hardware prototype is mandatory, demonstrated on a representative rotating or reciprocating machine - a bench-scale rig such as a motor with an inducible fault is acceptable, provided the fault mode is clearly stated. The solution must attach to existing equipment without major modification or extended downtime. The primary metric is prediction lead time - how far ahead of the failure the warning is raised - with detection accuracy and false-alarm rate as the counterweights. Teams must identify the specific failure mode or hazard targeted and justify sensor selection and	Manufacturing	

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must explain how false positives and false negatives are managed, since nuisance alarms destroy operator trust faster than missed faults. A retrofit bill of materials with per-machine installed cost is required to demonstrate viability for budget-constrained facilities.

41	Self-Diagnosing Edge Devices	Design and implement an embedded system capable of monitoring its own operational health and autonomously detecting degradation or fault conditions - such as sensor drift, abnormal power draw, communication errors, or thermal anomalies - without relying on continuous human monitoring or cloud connectivity. Faults are to be induced on the running device, by heating the board, injecting noise into a sensor line, or interrupting a bus, and caught while it continues to operate. Teams must state which health signals the device watches, how it identifies anomalies within the memory and compute headroom left on the chosen part, and what it does once a fault is confirmed: log it, degrade gracefully, or raise an alert. Characterise the cost of the self-diagnosis logic as flashed - bytes of flash and RAM occupied, CPU duty cycle, and incremental current draw measured at the supply rail - and justify why it remains viable on a cost-sensitive part. Teams	Digital Platforms

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		diagnostic reports are aggregated across a deployed fleet.		
42	Safeguarding Content Rights in the Age of AI-Generated Media	Content teams across an organization now use a wide range of generative models and tools, often without a consistent review gate before publication. Design a solution that assesses a media asset prior to deployment and reports on three aspects/dimension: • Provenance: whether the asset is AI-generated, wholly or partially, including tampering with existing provenance metadata or watermarks. • Rights exposure: likelihood that the asset reproduces copyrighted, trademarked or licensed material, including likeness and voice. • Deployment readiness: conformance to a configurable policy set covering brand, regulatory and regional requirements, with a clear go/no-go, flag or block outcome and the evidence behind it.	Cyber Security	Ask e Subn
43	GenAI-Powered Intelligent Manufacturing Knowledge Assistant for Shop Floor Operations	Develop a multimodal manufacturing knowledge assistant for one production line using a governed corpus of SOPs, work instructions, annotated equipment images, quality alerts, and maintenance/RCA reports. The system shall answer operator questions, retrieve procedures from images or text, and draft a change-controlled work	Manufacturing	Ask e Subn

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		must cite the exact source document, page/section, image region where applicable, and revision date. It must detect superseded documents, abstain when evidence is missing or conflicting, and route high-risk answers for expert approval. Compare with keyword search and a basic text-only RAG baseline on a fixed expert-authored question set.		
44	AI-Powered Threat Modeling & Security Validation for Embedded Systems	Design an AI-assisted threat modeling & security validation tool that: Automatically identifies attack surfaces from architecture diagrams/code Generates threat models (STRIDE, attack trees) Suggests security test cases and validation strategies	Cyber Security	
45	Governed Autonomous Multi-Agent Investigation System with Secure Tool Use	Build a governed multi-agent system in which a supervisor breaks an investigation into smaller jobs and issues each worker a formal work order defining its goal, success criteria, permitted tools, forbidden actions, resource limits, required evidence and stopping conditions. Each worker returns a standard evidence pack: its claims, the evidence for and against each, its assumptions, its confidence, a log of tools used, and whether it completed the job. Anything a tool returns must be treated as untrusted input and must never change a worker's goal, powers, permissions, budget or safety rules. Reading data and	Cyber Security	

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different levels of permission. The system must enforce resource budgets, prevent the same data-changing action running twice, handle failures gracefully, and return a partial answer when a job cannot be completed. The supervisor must compare findings and show where workers disagree rather than just picking one, and every order, tool call, decision and stop reason must be recorded in a permanent log.

46	Vehicle-to-Vehicle (V2V) Connectivity for Off-highway vehicles	Develop and demonstrate a POC in which at least two simulated or physical off-highway vehicles share information to improve safety or coordination. Participants may choose the operating environment, vehicle type, communication method, positioning approach, hardware, software, exchanged data, and user interface. The system should turn shared information into a warning, recommendation, or coordination action for a selected scenario, such as a blind crossing, sudden stop, reversing movement, restricted-zone entry, route conflict, queue, or another situation.The POC should be evaluated using suitable measures such as communication range, delay, message success, position accuracy, alert accuracy, response time, security, cost, ease of use.	Mobility	

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		Build a system that goes from an instruction to a completed physical task as one connected pipeline, rather than a set of separate AI system/sub-system bolted together. Given an instruction, it should identify the relevant objects or work area, understand how they are positioned relative to each other, work out the right sequence of moves, and carry out the task safely using one or more actuators. Solution must cope with normal real-world variation - objects in different positions and orientations, changing light, clutter and obstacles - without being reprogrammed each time. The target is a working prototype, or a simulation, that handles tasks it was not specifically set up for, corrects itself while running, operates safely near people, and clearly outperforms a traditional rule-based setup.		
47	Autonomous Vision-Language-Action Physical Systems		Robotics & Autonomous Systems	Ask e Subn
48	AI-Driven Battery Intelligence for Predictive Health, Safety, and Lifecycle Optimization	Develop an AI/ML proof of concept for one defined battery chemistry, cell or pack type, and application using public, synthetic data. The POC should preprocess voltage, current, temperature, charge-discharge cycles, capacity, and impedance or operating-condition data. It should estimate State of Health, predict Remaining Useful Life over a defined horizon, and detect selected abnormal thermal or electrical patterns. Every prediction must include confidence or uncertainty and	Energy	Ask e Subn

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		contributing signals. A dashboard should show battery history, current health, degradation trend, predicted end-of-life range, anomaly alerts, and model performance. Teams should compare the AI approach with simple capacity-threshold or trend-based baselines.		
49	Unified Multimodal Evidence Representation with Provenance, Temporal Alignment and Conflict Detection	Build an engine that reads information arriving in several different forms - such as text, images, tables or sensor readings over time - and turns it into one structured, inspectable picture per case: what is involved, what happened, what was measured, in what units, at what time, and how it all relates. Every fact must be stored with a pointer back to its exact source location, along with when it applies, how confident the system is, and whether it was read directly or worked out. The engine should recognise the same real-world item across different sources even when it is named differently, place events on a shared timeline, and flag a conflict only when two facts genuinely describe the same thing at the same time in comparable terms. Conflicts must stay visible, never averaged away. Teams should check their confidence scores against known answers and show how the engine behaves when a source is missing, a	AI	

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		or two sources disagree.		
50	Evidence-Grounded Divergence and Convergence Engine for Competing Hypotheses	Build a reasoning engine that works in two stages. First, it generates a small set of genuinely different candidate explanations for a problem, plus an explicit "unknown" option. They must differ in the cause proposed, the assumptions made, the evidence required or the effects predicted - not merely in wording. For each one, the engine lists what evidence supports it, what contradicts it, what is missing, and which single test would best separate it from the others. Second, it scores the candidates using a documented method and updates those scores as evidence arrives, without counting two pieces of evidence from the same original source as independent confirmation. The engine must end with one of four outcomes: pick the best-supported explanation, combine compatible ones, recommend a specific test, or openly decline to answer. It must report an honest confidence level and work within a fixed exploration budget with a clear rule for when to stop.	AI	

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